

Heart of Algebra

ANSWER KEY



Solving linear equations

- 1 Solve for x : $4x - 9 = 23$.

$$4x = 32, \text{ so } x = 8.$$

- 2 Solve for x : $\frac{3x + 2}{5} = 4$.

Multiply both sides by 5: $3x + 2 = 20$, so $3x = 18$, giving $x = 6$.

- 3 Solve for x : $6(x - 3) = 2(x + 9)$.

Expand: $6x - 18 = 2x + 18$. Subtract $2x$: $4x - 18 = 18$, so $4x = 36$, giving $x = 9$.

Linear inequalities

- 4 Solve $5x - 3 > 12$.

$$5x > 15, \text{ so } x > 3.$$

- 5 Solve $-2x + 7 \leq 15$.

Subtract 7: $-2x \leq 8$. Divide by -2 and flip the inequality: $x \geq -4$.

- 6 Solve $3(x - 2) \geq x + 4$.

Expand: $3x - 6 \geq x + 4$. Subtract x : $2x - 6 \geq 4$, so $2x \geq 10$, giving $x \geq 5$.

Slope and equations of lines

- 7 Find the slope of the line through $(2, 5)$ and $(6, 13)$.

$$m = \frac{13 - 5}{6 - 2} = \frac{8}{4} = 2.$$

- 8 Write the equation, in slope-intercept form, of the line with slope -3 passing through $(0, 7)$.

$$y = -3x + 7.$$

- 9 Find the slope of a line perpendicular to $y = \frac{2}{5}x - 1$.

A perpendicular slope is the negative reciprocal of $\frac{2}{5}$, which is $-\frac{5}{2}$.

Systems of equations by substitution

- 10** Solve the system: $y = 3x - 4$ and $2x + y = 11$.

Substitute: $2x + (3x - 4) = 11$, so $5x - 4 = 11$, giving $5x = 15$, $x = 3$. Then $y = 3(3) - 4 = 5$. Solution: $(3, 5)$.

- 11** Solve the system: $x - 2y = 1$ and $3x + y = 17$.

From the first equation, $x = 1 + 2y$. Substitute: $3(1 + 2y) + y = 17$, so $3 + 6y + y = 17$, giving $7y = 14$, $y = 2$. Then $x = 1 + 2(2) = 5$. Solution: $(5, 2)$.

Systems of equations by elimination

- 12** Solve the system: $3x + 2y = 19$ and $x - 2y = 1$, using elimination.

Adding the two equations eliminates y : $4x = 20$, so $x = 5$. Substituting into $x - 2y = 1$: $5 - 2y = 1$, so $2y = 4$, giving $y = 2$. Solution: $(5, 2)$.

- 13** Solve the system: $4x - y = 10$ and $2x + y = 14$, using elimination.

Adding the two equations eliminates y : $6x = 24$, so $x = 4$. Substituting into $2x + y = 14$: $8 + y = 14$, so $y = 6$. Solution: $(4, 6)$.

Absolute value equations

- 14** Solve $|2x - 5| = 9$.

$2x - 5 = 9 \Rightarrow x = 7$, or $2x - 5 = -9 \Rightarrow x = -2$. Solutions: $x = 7$ or $x = -2$.

- 15** Solve $|x + 4| = 10$.

$x + 4 = 10 \Rightarrow x = 6$, or $x + 4 = -10 \Rightarrow x = -14$. Solutions: $x = 6$ or $x = -14$.

Interpreting linear models

- 16** A phone plan costs $C = 20 + 0.05m$ dollars for m minutes of calls. Interpret the meaning of the numbers 20 and 0.05 in this context.

The 20 represents a fixed monthly fee (the cost when $m = 0$ minutes are used), and the 0.05 represents the cost per minute — each additional minute of calls adds 0.05 dollars to the bill.

- 17 Company A charges $C = 30 + 0.20d$ and Company B charges $C = 0.35d$ for a d -mile trip. Find the mileage at which both companies charge the same amount, and state which company is cheaper for a 150-mile trip.

Set equal: $30 + 0.2d = 0.35d$, so $30 = 0.15d$, giving $d = 200$ miles. At 150 miles: Company A charges $30 + 0.2(150) = 60$ dollars; Company B charges $0.35(150) = 52.5$ dollars. Company B is cheaper at 150 miles.

Systems with no solution or infinitely many solutions

- 18 For what value of k does the equation $6x + 4 = kx + 4$ have infinitely many solutions?

For infinitely many solutions the equation must hold for every x , so the coefficients of x must match: $k = 6$ (the constants already agree, $4 = 4$).

- 19 Explain why the system $2x - 3y = 6$ and $4x - 6y = 10$ has no solution.

Multiplying the first equation by 2 gives $4x - 6y = 12$. This contradicts the second equation, $4x - 6y = 10$ — the same left-hand expression cannot equal two different values, so the lines are parallel and never intersect.

Compound inequalities

- 20 Solve the compound inequality $-3 < 2x + 1 \leq 9$.

Subtract 1 from all three parts: $-4 < 2x \leq 8$. Divide by 2: $-2 < x \leq 4$.

- 21 Solve $4 - 3x \geq 13$.

Subtract 4: $-3x \geq 9$. Divide by -3 and flip the inequality: $x \leq -3$.

Multi-step word problems

- 22 Two numbers have a sum of 50 and a difference of 12. Find the two numbers.

Let the numbers be x and y with $x + y = 50$ and $x - y = 12$. Adding: $2x = 62$, so $x = 31$. Then $y = 50 - 31 = 19$. The numbers are 31 and 19.

- 23 The sum of three times a number and 7 equals 5 less than twice the number. Find the number.

$3x + 7 = 2x - 5$. Subtract $2x$: $x + 7 = -5$, so $x = -12$.

A well-designed word problem: ticket sales at a community theater

- 24** This question has three parts. A community theater sells adult tickets for 15 dollars and child tickets for 8 dollars. On Saturday the theater sold 120 tickets total and collected 1,450 dollars. (a) Set up a system of two equations for the number of adult tickets a and child tickets c sold. (b) Solve the system to find how many of each ticket type were sold. (c) On Sunday the theater raises the adult price to 18 dollars but keeps the child price the same. If the same number of adult and child tickets sell as on Saturday, find the new total revenue.

(a) $a + c = 120$ and $15a + 8c = 1450$. (b) From the first equation, $a = 120 - c$. Substitute:
 $15(120 - c) + 8c = 1450$, so $1800 - 15c + 8c = 1450$, giving $1800 - 7c = 1450$, so $7c = 350$,
 $c = 50$. Then $a = 120 - 50 = 70$. So 70 adult tickets and 50 child tickets were sold. (c) New revenue
 $= 18(70) + 8(50) = 1260 + 400 = 1660$ dollars.

Multiple choice — easy

- 25** Solve for x : $-2x + 19 = 23$.

- a. -4
- b. 0
- c. -2 ✓
- d. 2

$$-2x = 4, \text{ so } x = -2.$$

- 26** Solve for x : $-3x - 19 = 14$.

- a. 11
- b. -13
- c. -9
- d. -11 ✓

$$-3x = 33, \text{ so } x = -11.$$

27 Solve for x : $2x + 11 = 13$.

a. -1

b. 3

c. 4

d. 1 ✓

$$2x = 2, \text{ so } x = 1.$$

28 Solve for x : $7x + 15 = 29$.

a. 0

b. 2 ✓

c. -2

d. 4

$$7x = 14, \text{ so } x = 2.$$

29 Solve for x : $4x - 14 = 2$.

a. 6

b. -4

c. 4 ✓

d. 2

$$4x = 16, \text{ so } x = 4.$$

30 Solve for x : $6x - 9 = -51$.

a. -7 ✓

b. -5

c. -9

d. 7

$$6x = -42, \text{ so } x = -7.$$

31 Solve for x : $6x - 2 = 16$.

a. 5

b. 3 ✓

c. -3

d. 1

$6x = 18$, so $x = 3$.

32 Solve for x : $-9x + 20 = 119$.

a. -11 ✓

b. -13

c. -9

d. 11

$-9x = 99$, so $x = -11$.

33 Solve for x : $-3x - 19 = 26$.

a. -17

b. -15 ✓

c. 15

d. -13

$-3x = 45$, so $x = -15$.

34 Solve for x : $-8x + 1 = 9$.

- a. 2
- b. -3
- c. 1
- d. -1 ✓

$$-8x = 8, \text{ so } x = -1.$$

35 Solve for x : $-6x + 3 = -33$.

- a. 8
- b. 4
- c. 6 ✓
- d. -6

$$-6x = -36, \text{ so } x = 6.$$

36 Solve for x : $8x - 4 = -124$.

- a. -17
- b. 15
- c. -15 ✓
- d. -13

$$8x = -120, \text{ so } x = -15.$$

37 Solve for x : $-3x + 3 = -12$.

- a. 7
- b. 5 ✓
- c. -5
- d. 3

$$-3x = -15, \text{ so } x = 5.$$

38 Solve for x : $4x - 11 = -15$.

a. 2

b. 1

c. -1 ✓

d. -3

$4x = -4$, so $x = -1$.

39 Solve for x : $8x - 7 = 89$.

a. 14

b. 12 ✓

c. -12

d. 10

$8x = 96$, so $x = 12$.

Multiple choice — medium

40 Solve the system: $4x + y = -14$ and $4x - y = 6$, using elimination.

a. $(-1, -12)$

b. $(-1, -10)$ ✓

c. $(1, -10)$

d. $(-10, -1)$

Adding the two equations eliminates y : $8x = -8$, so $x = -1$. Substituting into the first equation:
 $4(-1) + y = -14$, so $y = -10$. Solution: $(-1, -10)$.

41 Solve the system: $5x - 5y = -55$ and $5x + 5y = 5$, using elimination.

a. $(-5, 6)$ ✓

b. $(6, -5)$

c. $(-3, 6)$

d. $(-5, 4)$

Adding the two equations eliminates y : $10x = -50$, so $x = -5$. Substituting into the first equation: $5(-5) - 5y = -55$, so $-5y = -30$, giving $y = 6$. Solution: $(-5, 6)$.

42 Solve the system: $2x + 5y = -1$ and $2x - 5y = -11$, using elimination.

a. $(-3, 1)$ ✓

b. $(-3, -1)$

c. $(-1, 1)$

d. $(1, -3)$

Adding the two equations eliminates y : $4x = -12$, so $x = -3$. Substituting into the first equation: $2(-3) + 5y = -1$, so $5y = 5$, giving $y = 1$. Solution: $(-3, 1)$.

43 Solve the system: $6x + 4y = -44$ and $6x - 4y = 20$, using elimination.

a. $(-2, -10)$

b. $(-8, -2)$

c. $(-2, -8)$ ✓

d. $(0, -8)$

Adding the two equations eliminates y : $12x = -24$, so $x = -2$. Substituting into the first equation: $6(-2) + 4y = -44$, so $4y = -32$, giving $y = -8$. Solution: $(-2, -8)$.

44 Solve the system: $3x - 3y = -21$ and $3x + 3y = 15$, using elimination.

a. $(-1, 4)$

b. $(-1, 6)$ ✓

c. $(1, 6)$

d. $(6, -1)$

Adding the two equations eliminates y : $6x = -6$, so $x = -1$. Substituting into the first equation: $3(-1) - 3y = -21$, so $-3y = -18$, giving $y = 6$. Solution: $(-1, 6)$.

45 Solve the system: $2x + 4y = -6$ and $2x - 4y = 18$, using elimination.

a. $(-3, 3)$

b. $(5, -3)$

c. $(3, -5)$

d. $(3, -3)$ ✓

Adding the two equations eliminates y : $4x = 12$, so $x = 3$. Substituting into the first equation: $2(3) + 4y = -6$, so $4y = -12$, giving $y = -3$. Solution: $(3, -3)$.

46 Solve the system: $4x - 5y = 1$ and $4x + 5y = 71$, using elimination.

a. $(9, 7)$ ✓

b. $(11, 7)$

c. $(7, 9)$

d. $(9, 5)$

Adding the two equations eliminates y : $8x = 72$, so $x = 9$. Substituting into the first equation: $4(9) - 5y = 1$, so $-5y = -35$, giving $y = 7$. Solution: $(9, 7)$.

47 Solve the system: $2x + 4y = 40$ and $2x - 4y = -24$, using elimination.

a. $(8, 4)$

b. $(4, 6)$

c. $(4, 8)$ ✓

d. $(6, 8)$

Adding the two equations eliminates y : $4x = 16$, so $x = 4$. Substituting into the first equation:
 $2(4) + 4y = 40$, so $4y = 32$, giving $y = 8$. Solution: $(4, 8)$.

48 Solve the system: $x - 2y = 6$ and $x + 2y = 6$, using elimination.

a. $(6, -2)$

b. $(6, 0)$ ✓

c. $(8, 0)$

d. $(0, 6)$

Adding the two equations eliminates y : $2x = 12$, so $x = 6$. Substituting into the first equation:
 $1(6) - 2y = 6$, so $-2y = 0$, giving $y = 0$. Solution: $(6, 0)$.

49 Solve the system: $2x - 6y = 38$ and $2x + 6y = -70$, using elimination.

a. $(-8, -9)$ ✓

b. $(-6, -9)$

c. $(-9, -8)$

d. $(-8, -11)$

Adding the two equations eliminates y : $4x = -32$, so $x = -8$. Substituting into the first equation:
 $2(-8) - 6y = 38$, so $-6y = 54$, giving $y = -9$. Solution: $(-8, -9)$.

50 Solve the system: $3x + y = -17$ and $3x - y = -37$, using elimination.

- a. $(10, -9)$
- b. $(-7, 10)$
- c. $(-9, 10)$ ✓
- d. $(-9, 8)$

Adding the two equations eliminates y : $6x = -54$, so $x = -9$. Substituting into the first equation:
 $3(-9) + y = -17$, so $y = 10$. Solution: $(-9, 10)$.

51 Solve the system: $x - y = 17$ and $x + y = -1$, using elimination.

- a. $(-9, 8)$
- b. $(8, -11)$
- c. $(8, -9)$ ✓
- d. $(10, -9)$

Adding the two equations eliminates y : $2x = 16$, so $x = 8$. Substituting into the first equation:
 $1(8) - y = 17$, so $-y = 9$, giving $y = -9$. Solution: $(8, -9)$.

52 Solve the system: $x - 5y = -48$ and $x + 5y = 52$, using elimination.

- a. $(2, 8)$
- b. $(4, 10)$
- c. $(2, 10)$ ✓
- d. $(10, 2)$

Adding the two equations eliminates y : $2x = 4$, so $x = 2$. Substituting into the first equation:
 $1(2) - 5y = -48$, so $-5y = -50$, giving $y = 10$. Solution: $(2, 10)$.

53 Solve the system: $6x - y = -56$ and $6x + y = -40$, using elimination.

- a. $(-8, 6)$
- b. $(-8, 8)$ ✓
- c. $(8, -8)$
- d. $(-6, 8)$

Adding the two equations eliminates y : $12x = -96$, so $x = -8$. Substituting into the first equation: $6(-8) - y = -56$, so $-y = -8$, giving $y = 8$. Solution: $(-8, 8)$.

54 Solve the system: $5x - y = -24$ and $5x + y = -6$, using elimination.

- a. $(-1, 9)$
- b. $(-3, 7)$
- c. $(-3, 9)$ ✓
- d. $(9, -3)$

Adding the two equations eliminates y : $10x = -30$, so $x = -3$. Substituting into the first equation: $5(-3) - y = -24$, so $-y = -9$, giving $y = 9$. Solution: $(-3, 9)$.

Multiple choice — hard

55 Solve $|2x - 9| = 25$.

- a. $8, -17$
- b. 17
- c. $-8, 18$
- d. $-8, 17$ ✓

$2x - 9 = 25 \Rightarrow x = 17$, or $2x - 9 = -25 \Rightarrow x = -8$. Solutions: $x = 17$ or $x = -8$.

56 Solve $|2x - 12| = 26$.

- a. 19
- b. $-7, 19$ ✓
- c. $7, -19$
- d. $-7, 20$

$2x - 12 = 26 \Rightarrow x = 19$, or $2x - 12 = -26 \Rightarrow x = -7$. Solutions: $x = 19$ or $x = -7$.

57 Solve $|3x| = 18$.

- a. $6, -6$
- b. $-6, 6$ ✓
- c. $-6, 7$
- d. 6

$3x = 18 \Rightarrow x = 6$, or $3x = -18 \Rightarrow x = -6$. Solutions: $x = 6$ or $x = -6$.

58 Solve $|2x + 12| = 14$.

- a. 1
- b. $-13, 2$
- c. $-13, 1$ ✓
- d. $13, -1$

$2x + 12 = 14 \Rightarrow x = 1$, or $2x + 12 = -14 \Rightarrow x = -13$. Solutions: $x = 1$ or $x = -13$.

59 Solve $|2x - 7| = 21$.

- a. 14
- b. $7, -14$
- c. $-7, 14$ ✓
- d. $-7, 15$

$2x - 7 = 21 \Rightarrow x = 14$, or $2x - 7 = -21 \Rightarrow x = -7$. Solutions: $x = 14$ or $x = -7$.

60 Solve $-5x + 11 \geq 1$.

a. $x \leq 2$ ✓

b. $x \leq 0$

c. $x \geq 2$

d. $x \leq 4$

Subtract 11: $-5x \geq -10$. Divide by -5 and flip the inequality: $x \leq 2$.

61 Solve $-4x - 2 \geq -42$.

a. $x \geq 10$

b. $x \leq 10$ ✓

c. $x \leq 12$

d. $x \leq 8$

Subtract -2 : $-4x \geq -40$. Divide by -4 and flip the inequality: $x \leq 10$.

62 Solve $-8x + 11 \geq 11$.

a. $x \leq -2$

b. $x \geq 0$

c. $x \leq 2$

d. $x \leq 0$ ✓

Subtract 11: $-8x \geq 0$. Divide by -8 and flip the inequality: $x \leq 0$.

63 Solve $-7x - 3 \geq 53$.

a. $x \leq -8$ ✓

b. $x \leq -10$

c. $x \leq -6$

d. $x \geq -8$

Subtract -3 : $-7x \geq 56$. Divide by -7 and flip the inequality: $x \leq -8$.

64 Solve $-6x - 12 \geq 48$.

a. $x \leq -8$

b. $x \geq -10$

c. $x \leq -12$

d. $x \leq -10$ ✓

Subtract -12 : $-6x \geq 60$. Divide by -6 and flip the inequality: $x \leq -10$.

65 Two numbers have a sum of 63 and a difference of 9. Find the larger number.

a. 38

b. 36 ✓

c. 34

d. 27

Let the numbers be x and y with $x + y = 63$ and $x - y = 9$. Adding: $2x = 72$, so $x = 36$. Then $y = 63 - 36 = 27$. The larger number is 36.

66 Two numbers have a sum of 114 and a difference of 16. Find the larger number.

- a. 63
- b. 49
- c. 67
- d. 65 ✓

Let the numbers be x and y with $x + y = 114$ and $x - y = 16$. Adding: $2x = 130$, so $x = 65$. Then $y = 114 - 65 = 49$. The larger number is 65.

67 Two numbers have a sum of 31 and a difference of 7. Find the larger number.

- a. 17
- b. 12
- c. 19 ✓
- d. 21

Let the numbers be x and y with $x + y = 31$ and $x - y = 7$. Adding: $2x = 38$, so $x = 19$. Then $y = 31 - 19 = 12$. The larger number is 19.

68 Two numbers have a sum of 147 and a difference of 15. Find the larger number.

- a. 81 ✓
- b. 79
- c. 66
- d. 83

Let the numbers be x and y with $x + y = 147$ and $x - y = 15$. Adding: $2x = 162$, so $x = 81$. Then $y = 147 - 81 = 66$. The larger number is 81.

69 Two numbers have a sum of 146 and a difference of 4. Find the larger number.

a. 71

b. 75 ✓

c. 73

d. 77

Let the numbers be x and y with $x + y = 146$ and $x - y = 4$. Adding: $2x = 150$, so $x = 75$. Then $y = 146 - 75 = 71$. The larger number is 75.